

GravField: Live-coding Bodies through Mixed Reality

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Figure 1: GravField is a live-coding intercorporeal improvisation system within a collocated Mixed Reality (MR) environment. In this system, MR headset wearers collaboratively improvise using their body movements to alter auditory and augmented visual feedback, based on MR audiovisual affordances dynamically programmed by live coders. The figure showcases three actual performance screenshots from a third-person spectator view, each demonstrating a different MR audiovisual affordance: "The Rope" (left), "The Spring" (middle), and "The Magnetic Field" (right).

Abstract

GravField (short for "Gravitational Field") is a live-coding intercorporeal improvisation system within a collocated Mixed Reality (MR) environment. It explores the dynamic interplay between bodies, mediated through live-coded MR audiovisual affordances. Inspired by contact improvisation techniques, the system offers several MR audiovisual affordances that drive different modes of intercorporeal interactions. For example, "Spring" shapes spatial relations through proximity, "Rope" fosters bodily synchronization, and "Magnetic Field" captures emotional responses of attraction or repulsion. The landscape of these affordances, dynamically adjusted by live coders, actively influences participants' collective movements, which in turn alter the live music generation, creating a feedback loop. Drawing on Merleau-Ponty's phenomenology, GravField illuminates the potential for future somaesthetic interplay systems in holistic interactive MR environments.

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CCS Concepts

• **Human-centered computing** → Interaction design theory, concepts and paradigms; **Mixed / augmented reality**.

Keywords

Augmented Reality, Collocated Mixed Reality, Live Coding, Dance, Affordance, Exergame, Social Play, Intercorporeality, Contact Improvisation, Movement-Sound Interaction

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In Human-Computer Interaction, with recent increasing trends focused on the experiential body, researchers have explored various facets of technology-mediated improvisational bodily movements. Examples include live-coding movement-sound interaction [Françoise et al. 2022], spatial awareness in contact improvisation [Lin et al. 2024], dance in front of mixed reality mirrors [Zhou et al. 2023], and engagement with physical design elements [Karpashevich et al. 2018], robots [Gemeinboeck and Saunders 2018], and drones [Dong et al. 2024]. However, the domain of technology-mediated inter-bodily improvisation remains unexplored. This field

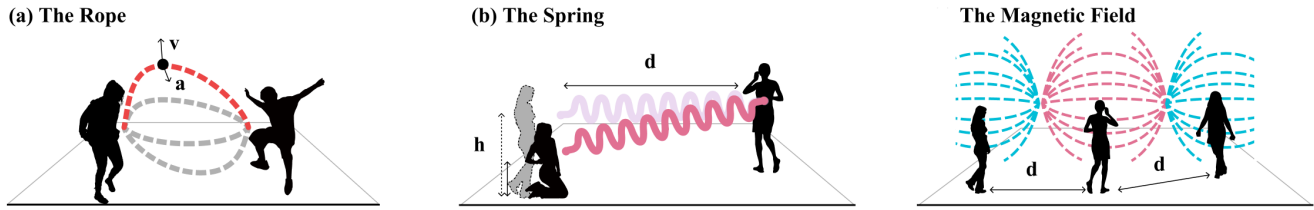


Figure 2: Live-coding MR audiovisual affordances: (a) **The Rope:** This affordance pattern explores proximity by linking auditory feedback to participants’ distance d and height h . Sound pitch and filter frequencies change to mirror the tension and compression of a spring, dynamically reflecting spatial relationships between participants; (b) **The Spring:** This affordance pattern examines synchronization by linking sound volume and pitch to the velocity v and acceleration a of simulated swinging motions. This setup mimics the experience of swinging a rope, enhancing the feeling of entrainment between participants; (c) **The Magnetic Field:** This affordance pattern investigates affinity by manipulating sound based on the distances d between participants. It creates a dynamic soundscape that reflects the forces of attraction and repulsion during the interaction.

draws upon Merleau-Ponty’s concept of intercorporeality, where perception and sense-making emerge through shared, intersubjective bodily interactions [Meyer et al. 2017]. Through these interactions, participants become aware of both their own bodies and those of others.

Inspired by contact improvisation techniques, we propose GravField, a live-coding intercorporeal improvisation system within a collocated Mixed Reality environment. GravField explores the dynamic interplay between bodies, mediated through virtual MR audiovisual affordances dynamically programmed by live coders (Figure 2). These MR affordances provide intuitive, imaginative guidance for participants’ collective movements, while the system’s visual and auditory feedback simultaneously responds to those movements.

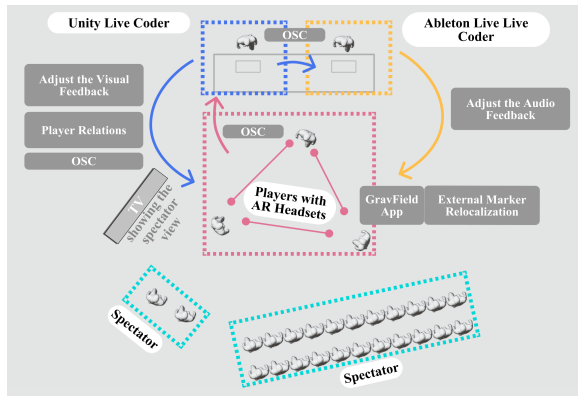


Figure 3: GravField performance layout

GravField’s implementation is based on InstantCopresence [Hu et al. 2023], a Collocated Multiplayer MR framework that synchronizes the spatial poses of all collocated devices within a local network. We use HoloKit [Hu et al. 2024], an open-source MR headset with stereoscopic displays, for all improvisational participants. Live coders employ Ableton Live for real-time sound synthesis and Unity’s Visual Effect Graph to create dynamic virtual MR affordances that respond to participants’ movements. The system adjusts auditory elements—pitch, volume, and texture—based on

participants’ positions and movements, reflecting the dynamics of proximity, synchronization, and affinity. This design enables live coders to continuously adjust the soundscape, maintaining a fluid and engaging interaction experience for participants.

GravField not only engages participants and live coders but also involves the audience through a third-person spectator view of the MR environment (Figure 1). This view allows spectators to observe MR interactions via collocated AR devices such as handheld iPads or fixed screens. Consequently, GravField’s performance setup (Figure 3) can accommodate a large audience as spectators.

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